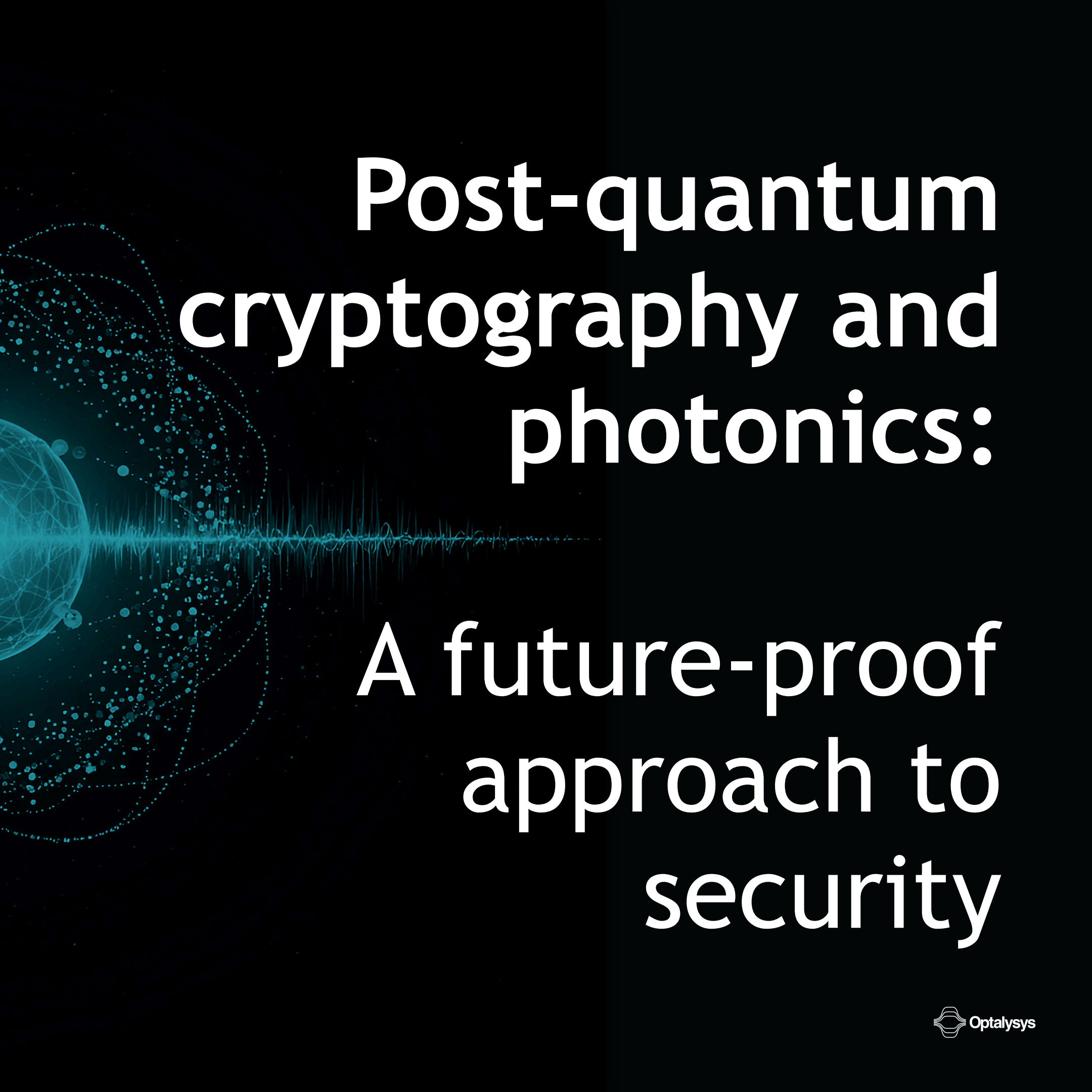




Post-quantum cryptography and photonics:

A future-proof approach to security

The quantum shift

Quantum computing is accelerating faster than expected, delivering processing leaps that threaten today's security foundations.

Why it matters

Quantum machines can solve problems in seconds that would take classical computers thousands of years, putting current encryption at risk.

A global security threat

Most of the world's digital infrastructure — from banking to healthcare to government — still relies on encryption algorithms vulnerable to quantum attacks.

Economic impact

A single large-scale quantum attack could trigger trillions in economic damage and systemic financial failures.

The threat is already here

Adversaries are stealing
encrypted data today to
decrypt once quantum
power becomes available —
“harvest now, decrypt later”

Countdown to Q-Day

Companies have moved their quantum-readiness timelines forward to 2029 due to accelerating breakthroughs.

Post-quantum cryptography

PQC provides a path forward... but comes with high computational cost that slows systems and increases energy demands.

Enter... photonics

Silicon photonics uses light, not electricity, to process data, delivering massive parallelism, minimal heat, and ultra-low energy use.

Why photonics changes everything

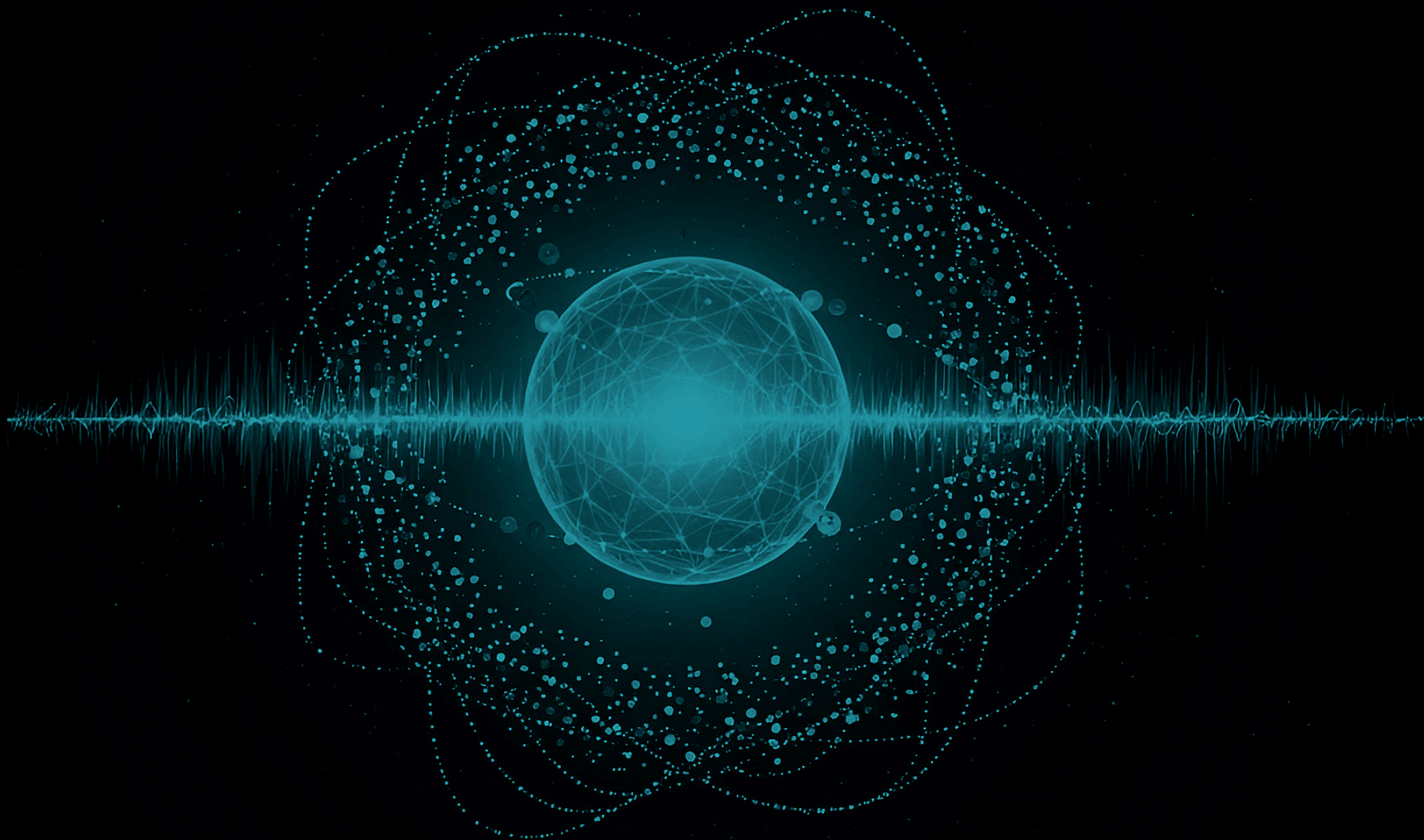
Photonic compute can
accelerate lattice-based
PQC workloads, cutting
latency and energy
consumption dramatically.

Future-proof security

This approach enables strong post-quantum protection, including FHE, without slowing performance or inflating budgets.

Photonics insights

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